

JAMSHEDPUR, India

WMO: 427980

Lat: 22.813N

Lon: 86.169E

Elev: 154

StdP: 99.49

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.1	(c) 11.2	(d) 1.6	(e) 4.3	(f) 28.2	(g) 3.6	(h) 5.0	(i) 26.8	(j) 3.5	(k) 22.9	(l) 3.1	(m) 23.1	(n) 0.1	(o) 270	(p) 0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 12.1	(c) 42.1	(d) 22.6	(e) 40.6	(f) 22.9	(g) 39.0	(h) 23.4	(i) 28.2	(j) 33.6	(k) 27.8	(l) 32.7	(m) 27.4	(n) 32.1	(o) 1.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	23.3	30.4	26.7	22.7	29.8	26.3	22.2	29.4	92.8	33.7	90.4	33.0	88.7	32.3	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.4	(b) 3.5	(c) 3.1	(e) 8.0	(f) 44.4	(g) 0.6	(h) 1.2	(i) 7.6	(j) 45.3	(k) 7.2	(l) 46.0	(m) 6.9	(n) 46.7	(o) 6.4	(p) 47.6		
(a) 4.4	(b) 3.5	(c) 3.1	(e) 8.0	(f) 44.4	(g) 0.6	(h) 1.2	(i) 7.6	(j) 45.3	(k) 7.2	(l) 46.0	(m) 6.9	(n) 46.7	(o) 6.4	(p) 47.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	18.9	22.8	27.6	31.9	32.4	31.4	29.1	28.8	28.7	27.1	23.3	19.5		
	DBStd	4.88	1.92	2.52	2.47	2.45	2.29	2.78	1.41	1.28	1.39	1.72	2.04	1.84		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	25	15	1	0	0	0	0	0	0	0	0	0	9		
	CDD10.0	6134	277	359	546	656	696	641	591	583	561	529	401	295		
	CDD18.3	3118	34	127	288	406	437	391	332	325	311	271	151	46		
	CDH23.3	37556	321	1206	3422	5829	6303	5374	3918	3748	3486	2580	1042	328		
(14) Wind	WSAvg	0.8	0.4	0.6	0.8	1.2	1.2	1.2	1.1	0.9	0.9	0.5	0.3	0.4		
	PrecAvg	1448	17	14	18	25	84	222	351	323	282	82	19	10		
	PrecMax	2020	133	87	99	124	176	641	746	594	488	291	138	96		
	PrecMin	642	0	0	0	0	1	32	96	129	113	14	4	0		
	PrecStd	308	27	21	25	32	49	135	147	120	104	71	27	22		
	DB	30.2	35.8	40.2	44.1	43.4	43.4	35.8	34.7	35.0	34.4	32.1	29.2			
	MCWB	17.8	19.2	20.8	21.1	22.4	24.2	27.3	27.1	26.1	24.1	22.0	18.2			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	28.2	33.3	38.0	42.2	41.7	41.1	34.2	33.6	33.8	33.2	30.7	27.8			
	2% MCWB	17.2	18.9	20.4	21.3	23.1	25.5	27.1	27.1	26.4	24.6	21.3	17.7			
	5% DB	26.6	31.5	36.3	40.7	40.2	38.6	33.1	32.6	32.9	32.2	29.7	26.6			
	5% MCWB	16.3	18.3	20.0	21.5	23.9	26.3	27.0	27.0	26.5	24.5	20.9	17.2			
	10% DB	25.1	29.7	34.8	39.1	38.6	36.5	32.0	31.7	32.0	31.2	28.6	25.3			
	10% MCWB	15.9	18.0	19.8	21.7	24.8	26.6	26.8	26.8	26.4	24.1	20.4	16.7			
	WB	20.1	23.1	26.2	27.3	28.8	29.2	28.4	28.2	27.9	27.0	24.1	21.1			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	25.2	28.9	32.8	33.0	34.8	35.0	33.0	32.2	32.2	31.2	28.5	25.8			
	2% WB	19.1	21.1	24.1	26.6	28.1	28.4	27.7	27.7	27.3	26.4	23.4	19.7			
	2% MCDB	24.0	27.4	30.9	32.4	33.9	34.3	31.9	31.6	31.4	30.3	27.5	24.3			
	5% WB	18.2	20.1	23.0	25.8	27.5	27.8	27.4	27.3	27.0	26.0	22.7	18.7			
	5% MCDB	23.5	26.7	30.1	32.0	33.6	33.5	31.2	30.9	30.9	29.7	26.8	23.8			
	10% WB	17.3	19.3	21.9	25.0	26.7	27.3	27.1	27.0	26.7	25.6	22.0	17.8			
	10% MCDB	23.1	26.5	29.6	32.0	33.4	32.9	30.7	30.5	30.3	29.1	26.2	23.0			
(35) Mean Daily Temperature Range	MDBR	13.8	13.9	14.5	14.2	12.1	8.7	5.7	5.7	6.4	8.7	11.9	13.4			
	5% DB MCDBR	15.7	16.1	16.3	16.1	14.1	12.3	7.5	7.1	8.0	9.8	12.2	14.3			
	5% WB MCWBR	5.9	4.9	4.7	4.7	4.3	3.6	2.4	2.2	2.3	3.0	4.1	5.3			
	5% DB MCDBR	12.4	12.7	13.7	14.0	12.0	9.8	6.3	6.2	6.8	7.6	9.5	11.8			
	5% WB MCWBR	5.4	5.0	5.4	5.4	5.0	3.5	2.4	2.3	2.4	2.7	3.3	4.9			
(40) Clear-Sky Solar Irradiance	taub	0.527	0.513	0.560	0.593	0.651	0.679	0.593	0.579	0.587	0.589	0.599	0.571			
	taud	1.845	1.917	1.839	1.814	1.739	1.707	1.905	1.955	1.917	1.853	1.748	1.778			
	Ebn at Noon	722	774	759	742	695	671	731	743	727	699	654	662			
	Edh at Noon	188	186	210	219	234	240	197	187	191	196	206	195			
(44) All-Sky Solar Radiation	RadAvg	3.99	4.88	5.78	6.41	6.35	5.09	4.13	4.25	4.38	4.53	4.11	3.72			
	RadStd	0.27	0.29	0.24	0.27	0.29	0.42	0.34	0.23	0.28	0.41	0.22	0.24			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	+0.33	+0.38	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page